Software Process Management

Lecture 4

Recap

- We discussed an overview of software, and their development
- We discussed the challenges in software development and lifecycle
- Also, we initiated the discussion on software processes and process models

Elements of Process Models

- A description of an identifiable activity or a group of activities
- A description of the product flow (i.e., input and output products for activities)
- A description of the control flow between processes (i.e., the enactment or execution sequence)

Elements of Process Models

- A description of a refinement (i.e., the definition of a hierarchy of processes)
- A description of the relationships to techniques, methods, tools
- A description of the relationship to roles

Terminology

- **Process description:**
- While a process is a vehicle for doing a job, a process description is a specification of how the job is to be done.

- Thus, cookbook recipes are process descriptions, while preparing a recipe is a process.

Terminology

- **Process model:**
- A software process model reflects an organization's know-how regarding software development. Software engineering know-how has to be developed and maintained. Practical experience has shown the need for modelling software engineering entities (especially processes), measuring those entities, reusing the models, and improving the models.

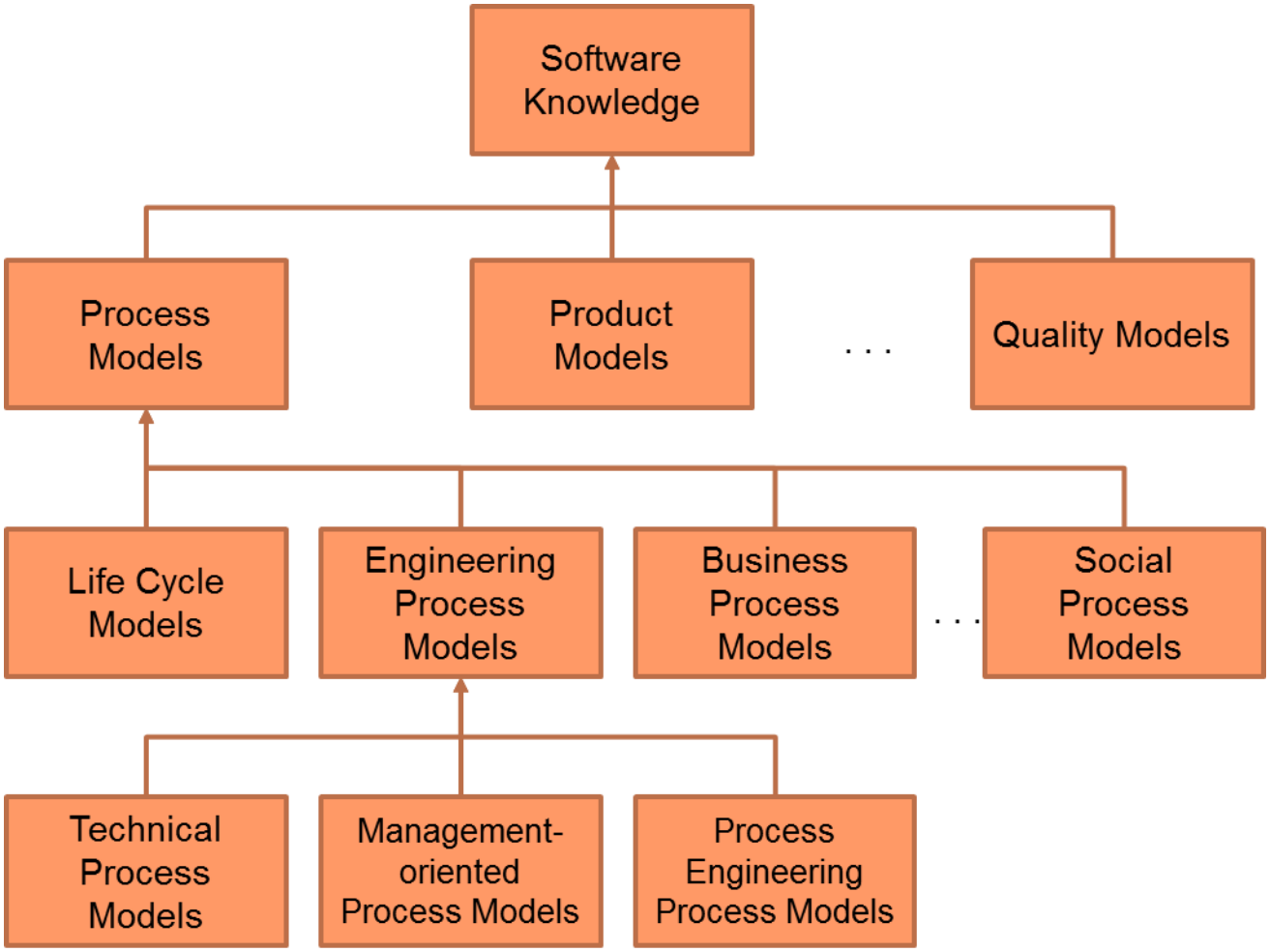
…And More Terminology

- **An atomic process** (synonym: process step) is a process that does not allow further structuring in the form of sub processes.

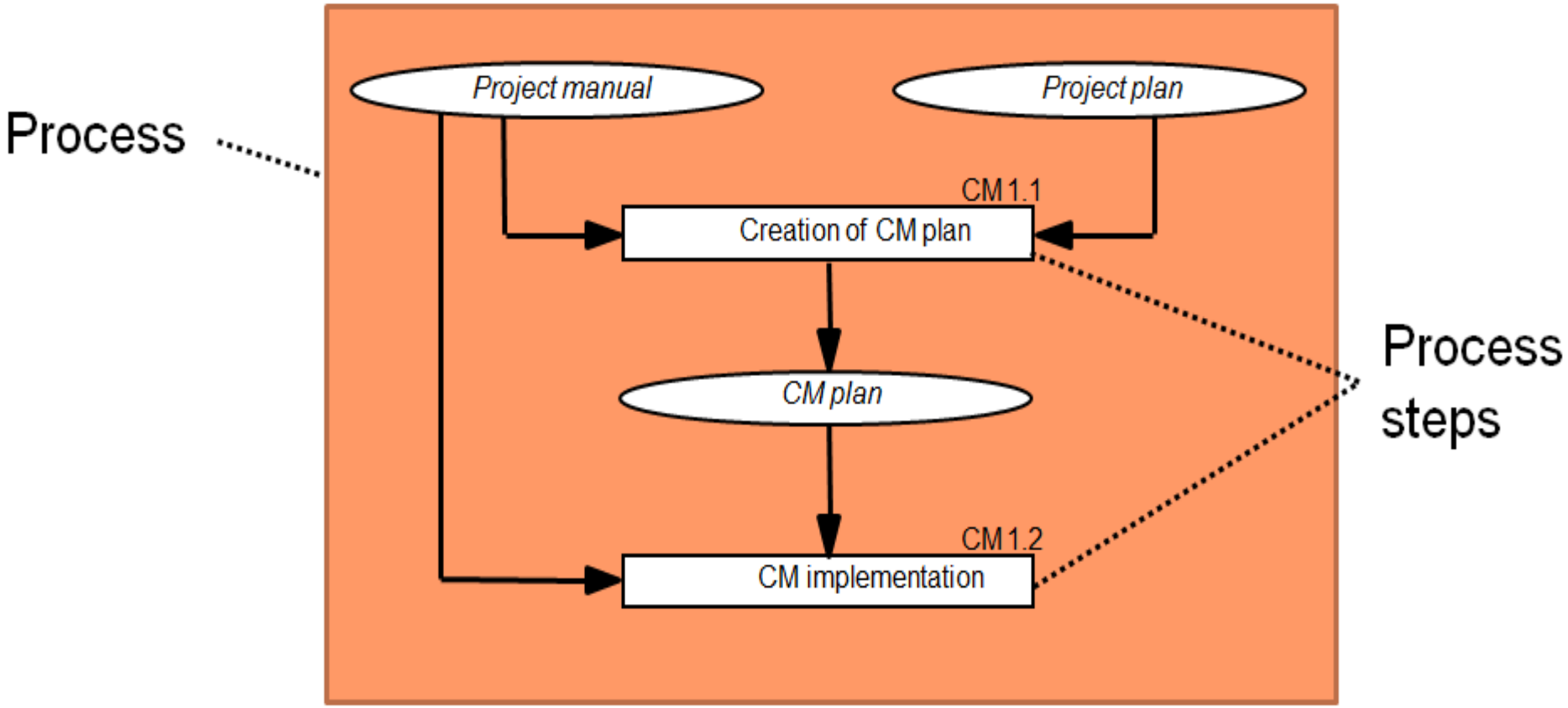
- **Process enactment** is the performance of process steps undertaken to reach a given goal. The process performer (i.e., “agent”) can be a human or a machine. In case of a machine, the term “process execution” is usually used.

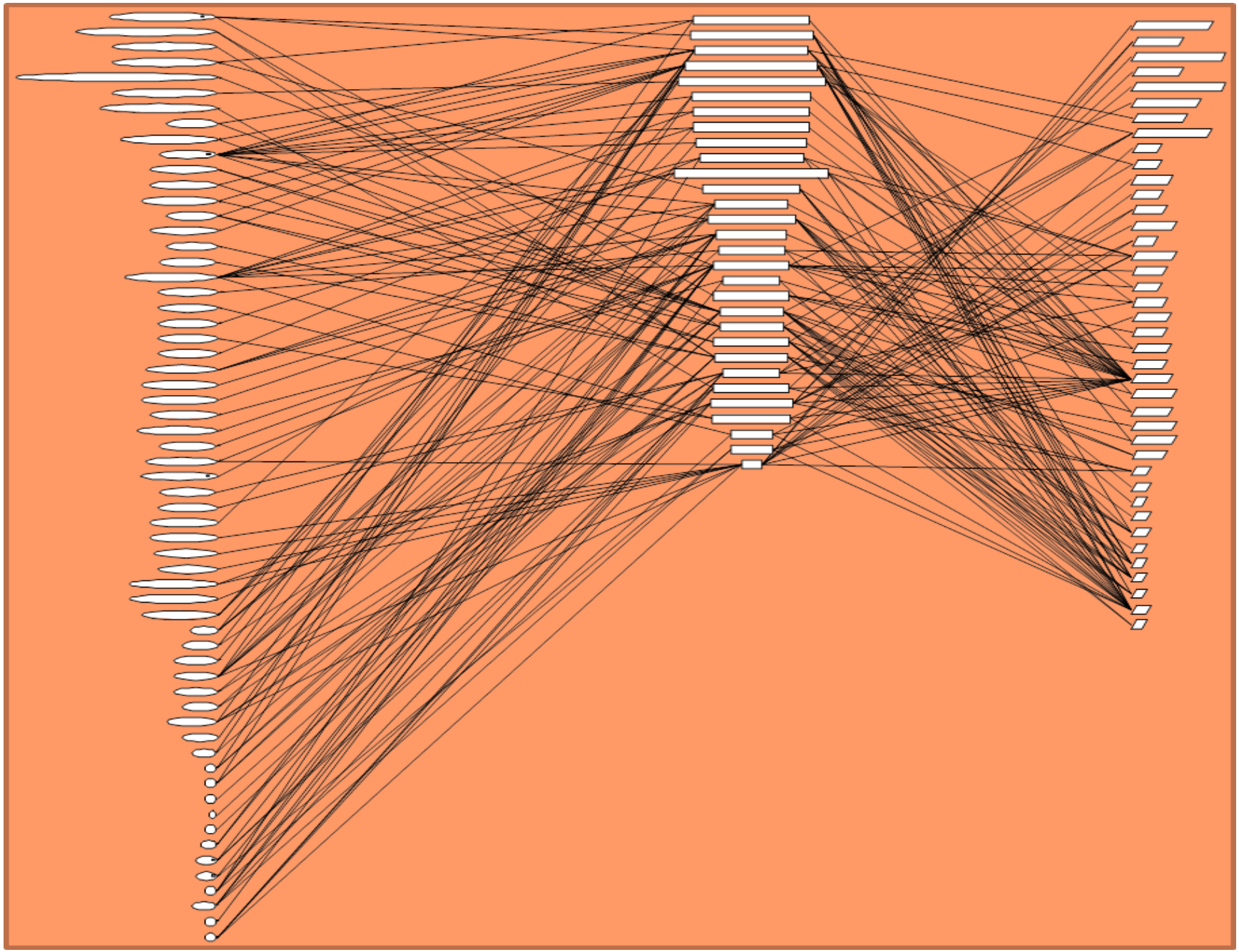
- **A process definition** is a description of a process that is enactable.

Hierarchy of Software Processes



Example


Process
Project manual
Project plan
CM 1.1
Creation of CM plan
CM plan
CM 1.2
CM implementation
Process steps



More terms….

- Process scripts and process programs are specializations of process definitions:

- **A process script** is a description of a process that is suitable for interpretation by humans.

- A process script should be tailored to the needs of the process performer.

Terms …

- **A process program** is a description of a process that can be interpreted by machines.

- **A process schema** (synonym: process meta-model, process architecture) is a conceptual framework for the consistent description of process models and their relationships.

- A process schema describes, on the one hand, building blocks and their relationships that form a process model, and, on the other hand, constraints on their composition.

Process Schema

- Only few process management tools are flexible enough to cope with multiple process schemata or are able to import individual process schemata.
- Often, a process schema is created ad hoc together with the process model.
- This often implies description failures (e.g., phases are refined by process models).

How to define a process

- Example sequence for the definition of processes (heuristic):
  - Identification of products and their refinements
  - Identification of processes and their refinements
  - Definition of product flows
  - Definition of control flow
  - Assignment of techniques / methods / tools
  - Generation of process documentation (e.g., web-based description, handbook, training material)
- This is a concept-oriented approach
- There are other approaches (e.g., role-oriented, abstraction-oriented, control-flow-oriented)

- **A process agent** (synonym: process performer) is a person or machine that enacts/executes the process in order to reach the process goal(s). Humans interpret process scripts, machines interpret process programs.

- **A process owner** is a human or organizational entity that sets the goals of a process and is responsible for their achievement.

- A process owner provides resources for the enactment or execution of the process and is responsible for providing appropriate process definitions.

- **A process engineer** is a person who pursues one or several goals of process modelling (e.g., defining, extending, maintaining, improving process models).

- To that end, a process engineer uses process models, which he defines, extends, improves, and manages.

- The process engineer should pay attention to the accuracy of the model, i.e., the correspondence between the real-world process enactment/ execution and the process model.

- A principle is a policy or mode of action that describes important characteristics of a process model.

- Often, new process models evolve from principles.

Examples of principles

- Active user involvement is imperative
- Frequent inspection
- Work in progress should be limited
- Time-boxed iterations
- Develop small incremental releases and iterate
- Frequent delivery of product
- Continuous integration
- Colocation
- Common coding guidelines
- Self-organizing teams
- Daily meeting

- A principle is not a process or a process model, but a process or a process model can capture one or more principles.
- Principles should be adapted to contexts (e.g., by using experimentation) and integrated into process models.
- Sometimes the term “practice” is used as a synonym for principle, e.g., the principle “continuous integration” is often referred to as an XP practice.

- A **product** is each artefact that is consumed or produced in the context of engineering-style software development.

- Products can be refined by other products

Examples of products

- Source code
- Specification document
- Problem description
- Configuration data
- Component design
- Test case
- Test result
- Project plan

- A **product model** is a description of a product or a class of products.

- Usually, software product models consist of a description of the information units of a software product
  - (e.g., functional requirements, non-functional requirements, design decisions)

- and a structure for arranging the information units
  - (e.g., a table of contents for a requirements document).

- The **product flow** consists of the relationships between products and processes that describe the access mode to the products.
- The following access modes are typically defined:
- – Produce (write)
- – Consume (read)
- – Modify (read/write)

- A **role** is a set of processes belonging together that are assigned to one or several agents. A role combines the functional responsibility for the enactment of a process.

Roles

- Examples of technical development roles are:

- Requirements engineer,
- Designer (architecture),
- Designer (data/algorithms),
- Programmer,
- Inspector,
- System integration engineer,
- Tester.

Roles

- Examples of organizational and management-oriented roles are
- Project planner
- Project manager
- Product manager
- and Quality engineer.

- Like a process, a role is an abstraction. A role groups competencies (i.e., knowledge and rights).
- Several different types of relationships between roles and agents can be defined, especially
- 1:1 (e.g., the tester is Mr. Miller),
- m:1 (e.g., Mr. Miller plays both the role of the requirements engineer and the role of the code inspector),
- 1:n (the role of the project manager is shared by Mr. Miller and Ms. Scott),
- m:n (a mixture of the previous cases).
- It is important that the relationship between a role and an agent is explicitly defined. This is typically done during project planning, resource planning, and re-planning.

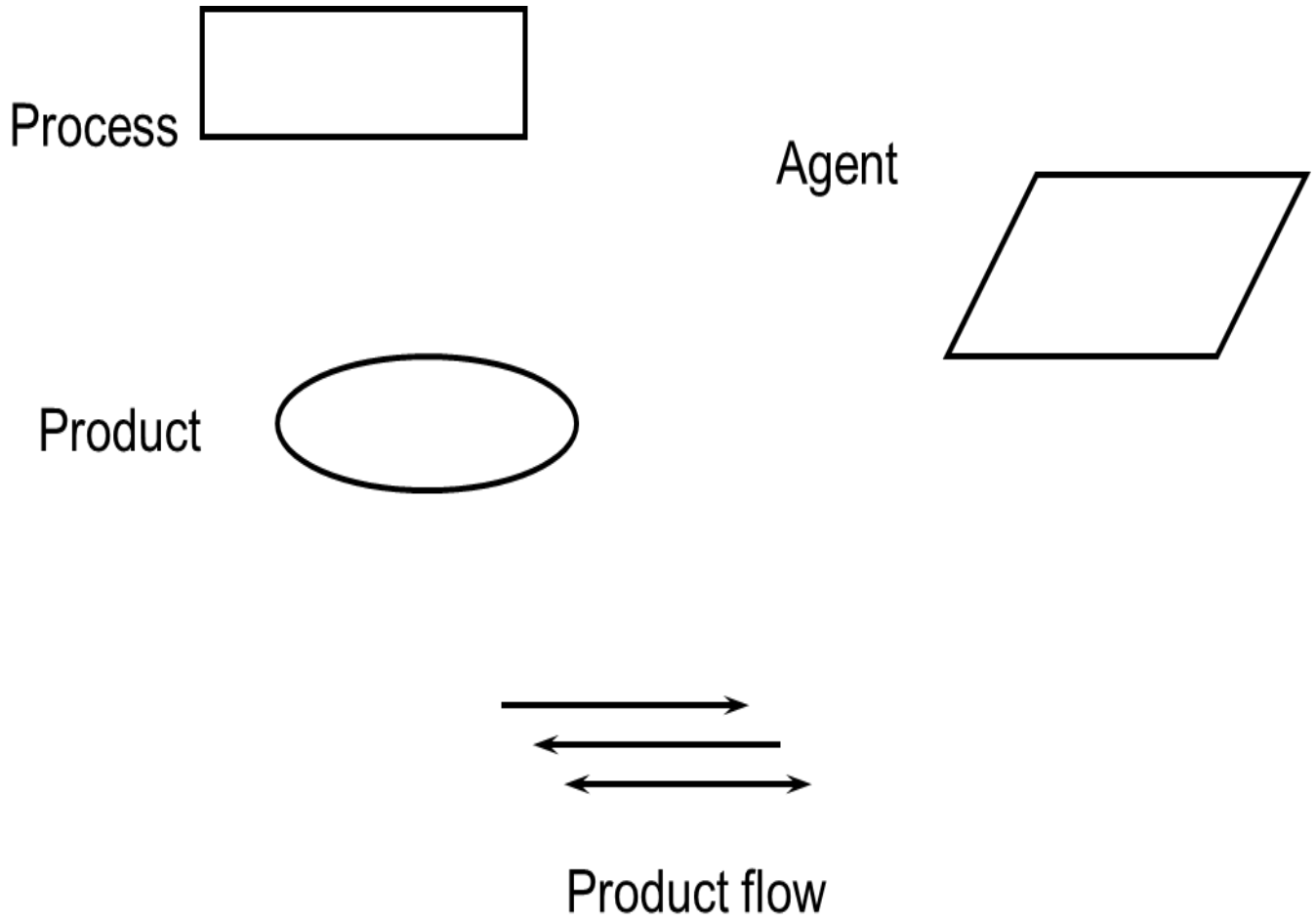
- A role is typically described as a specific view on the process by:

- – The activities the role is involved in (described independently of the person)
- – The documents that are consumed or produced in these activities
- – The level of involvement of a role in an activity
- – The requirements for playing a role (e.g., qualifications)

Selecting what to show

- Often, only selected elements are shown in graphical representations of process models.
- Figure 1.4, for instance, shows only activities, artefacts, and the product flow, while
- Fig. 1.5 also shows the control flow and the relationships to roles.

Example Graphical Representation (used e.g., in MVP-L)



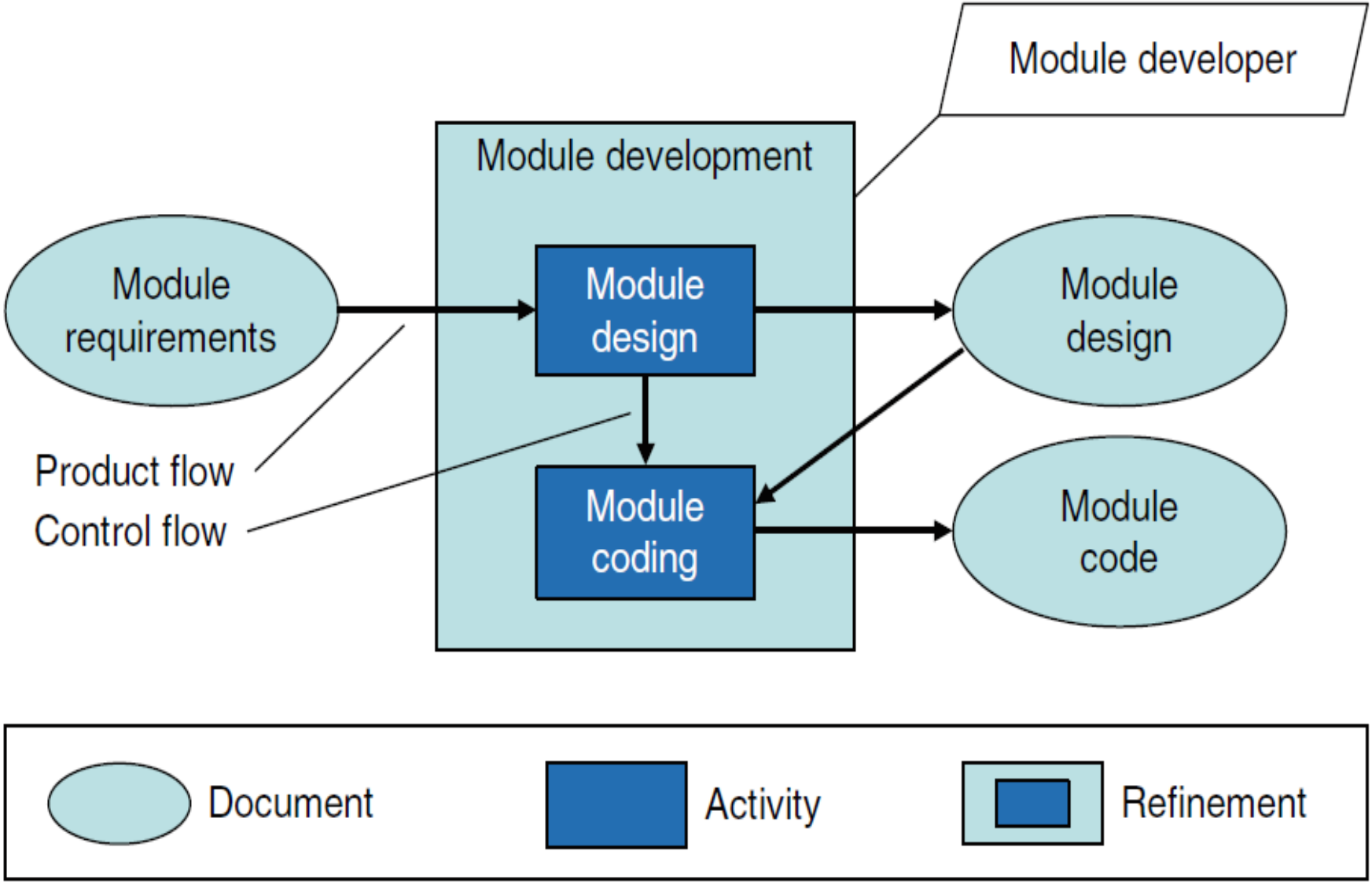


Fig. 1.6 Role example

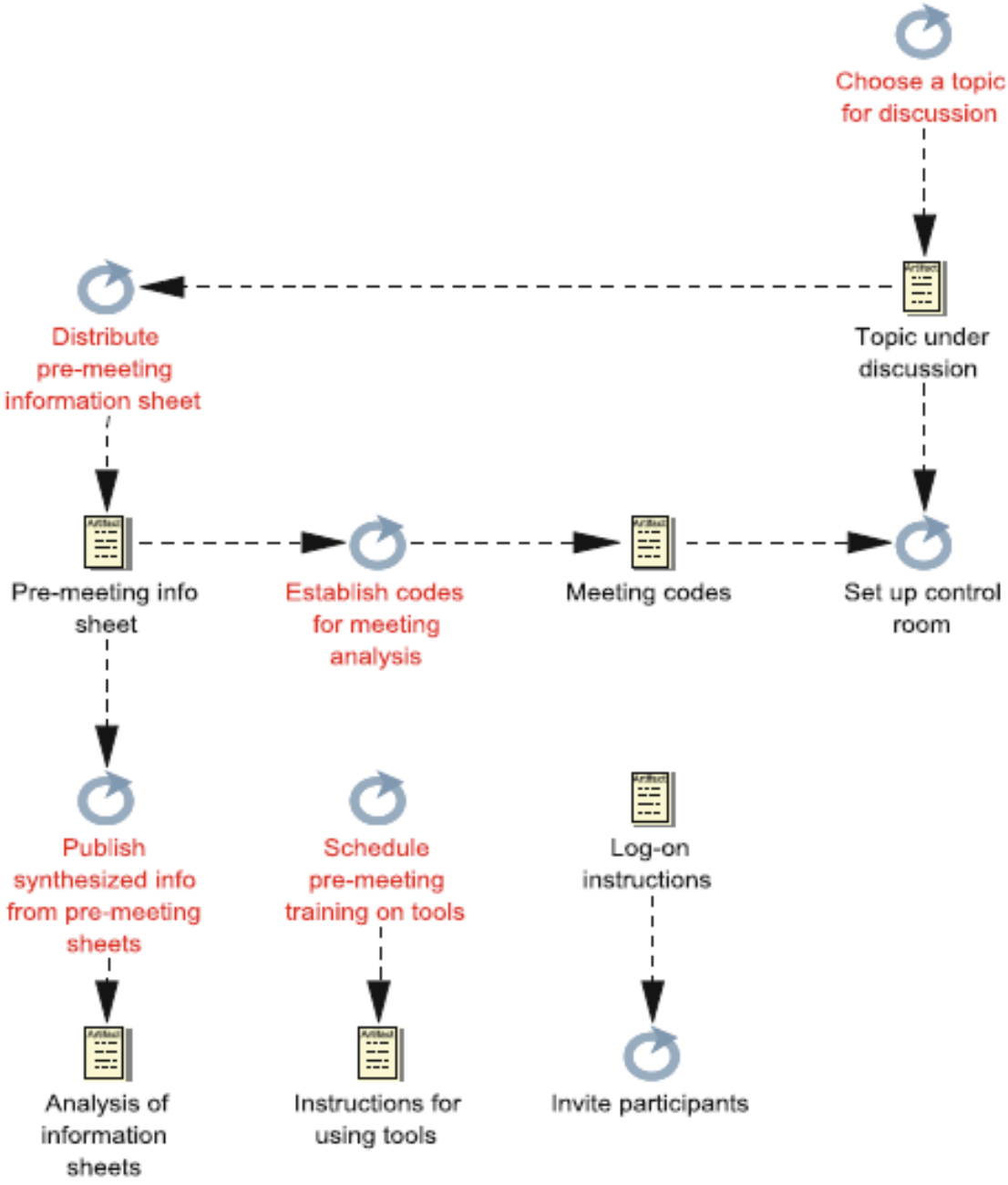
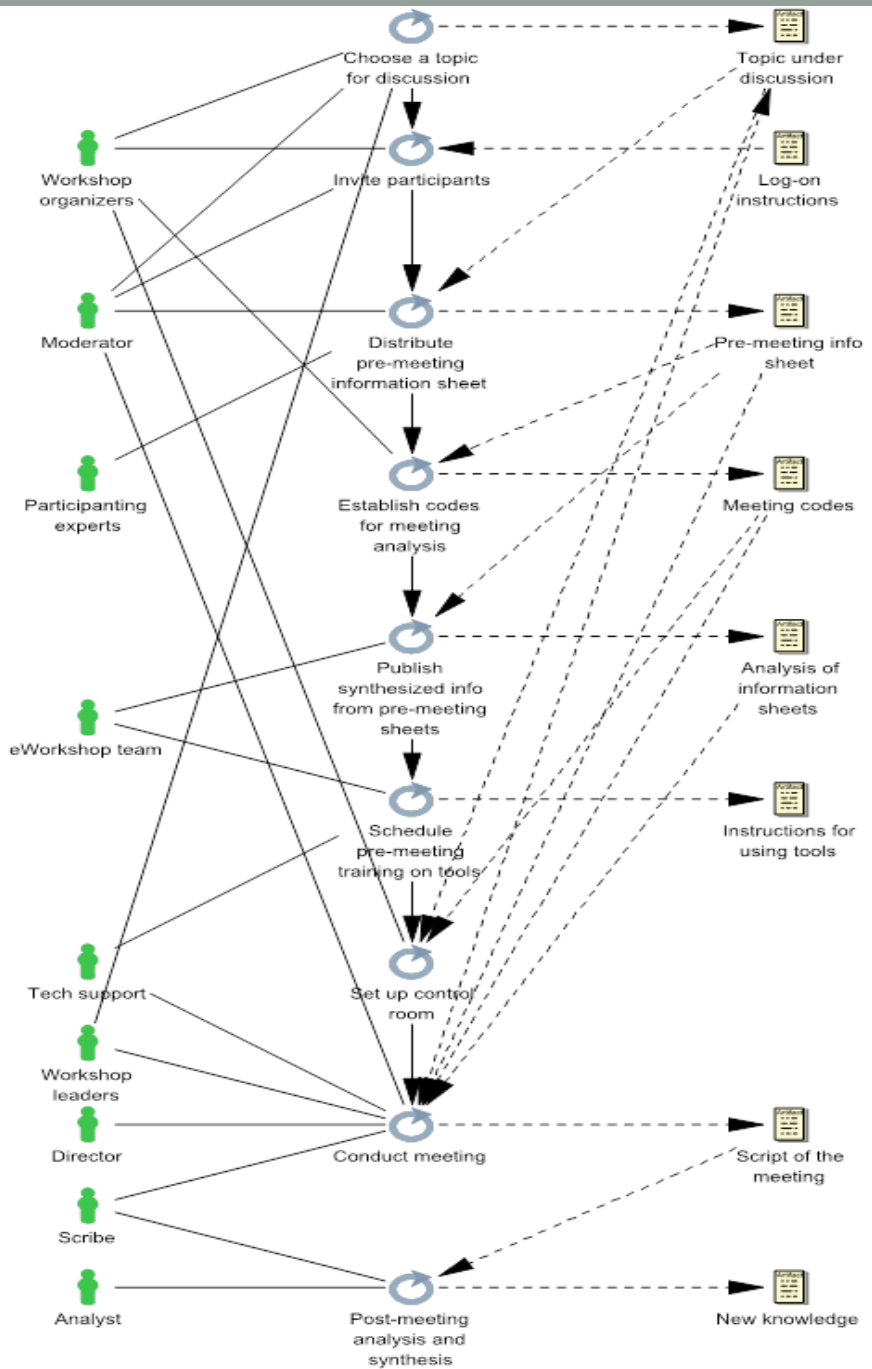


Fig. 1.4 Process model with product flow


Choose a topic for discussion
Topic under discussion
Workshop organizers
Invite participants
Log-on instructions
Moderator
Distribute pre-meeting information sheet
Pre-meeting info sheet
Participanting experts
Establish codes for meeting analysis
Meeting codes
Publish synthesized info from pre-meeting sheets
Analysis of information sheets
eWorkshop team
Schedule pre-meeting training on tools
Instructions for using tools
Tech support
Set up control room
Workshop leaders
Director
Conduct meeting
Script of the meeting
Scribe
Analyst
Post-meeting analysis and synthesis
New knowledge

Benefits of the role concept are:

- Activities and responsibilities of persons involved in a project can be clearly defined
- Transparency supports communication and cooperation between roles
- Necessary competencies are clearly defined
- Project planning is simplified
- Resources for a project or an organization can be specified independently of available personnel
- Role bottlenecks can already be determined at an early stage during project planning

Roles and Process Models

- The relationship between roles and process models can be qualified:
- For instance, a role can perform a process or a role needs to be informed when a process is performed or completed.
- Often, a so-called “Responsibility Assignment Matrix” (RAM) is used to qualify the relationship between roles and processes.

Responsibilities

- RASCI Matrix
  - R = Responsible [Process Owner]
  - A = Approval
  - S = Support [Process Agent]
  - C = Consulting
  - I = Informing

Roles / Activities	Module Developer	Moderator	Tester	Quality Assurance Manager
Module Development	R, S			
Module Coding	R, S			
Module Review	R, S	S		A
Module Testing	I		R, S	I

Example process script

- **“Modify Test Plans”**
- Description
  - This step involves the modification of test plans and objectives to include testing of capabilities related to the requirements change prompting this software modification. [...]
  - [The test plans] identify the functionality and capabilities to be tested, and the approach to be taken.
  - On the other hand, the specifics of actual test data, procedures, and so forth are dealt with in the subsequent step (modify unit test package).
- Inputs
  - Current test plans (from test plans file) [...]
- Outputs
  - Modified test plans (to test plans file) [...]
- Responsibility
  - This step is carried out by the assigned QA engineer.
- Constraints
  - This step can begin as soon as the task has been assigned by the project manager.
  - This step ends when its output has been provided.

Example of a process programme

```
TYPEPROCESS release
1 EVENT ready = ( state := ready ) ;
  ROLE USER  = PManager;
  ROLE implement = development;
  ROLE valid = validation;
  ROLE component = module ; { ON ready DO {
2 IF implement.to_change.%name.state == ready THEN
3  implement.to_change.%name.state := available;
...
```

- A **project plan** is a specification of
- the necessary resources for the execution of a process definition,
- the relationships between these resources and processes,
- the produced products including the product flows, and
- restrictions of any type concerning the execution of the process.

4 Domain Principle

